

PRE-DEPLOYMENT EVALUATION OF TARGETED AUTOMATIC SPEECH RECOGNITION ATTACKS OVER LIVEKIT CALLS

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ABSTRACT

Targeted automatic speech recognition (ASR) attacks can perform differently on local networks with and without added delay and jitter. We investigate whether calibrating local transport conditions to deployed measurements improves pre-deployment evaluation. We evaluate 72 adversarial recordings constructed against Whisper under optimization transformations, LiveKit Local baseline, Local calibrated, and LiveKit Cloud. Calibration reduced the mean absolute difference between Local and Cloud recording-level success rates from 11.1 to 7.4 percentage points. Optimization transformations achieved 51.6% exact-target success versus 21.9% through Cloud, with a mean absolute discrepancy of 30.7 percentage points. Our empirical results show that optimization-time success can overestimate deployed attack performance and that deployment-informed transport calibration can improve local evaluation in this setup.

Index Terms— Automatic speech recognition, adversarial audio, Voice over IP, WebRTC, network emulation

1. INTRODUCTION

A targeted adversarial attack on an automatic speech recognition (ASR) model modifies a recording so that the model produces an attacker-selected transcript. An attack that succeeds may fail after passing through a call path because the audio is resampled, encoded, packetized, delayed, affected by packet loss, reconstructed, and time-scaled. An over-the-call attack must therefore remain effective after both audio processing and network impairments. One approach is to add a small adversarial perturbation to the recording and optimize it under transformations that approximate effects introduced by the call path. The usefulness of those transformations depends on whether success under them corresponds to success after actual call delivery. It follows that if that correspondence is poor, then testing against optimization transformations can give a misleading estimate of deployment performance.

Carlini and Wagner demonstrated targeted white-box attacks against an end-to-end speech recognition system [1]. Athalye et al.’s expectation-over-transformation (EOT) method optimizes adversarial examples over a distribution of

transformations [2]. Qin et al. later optimized targeted audio against simulated environmental distortions and used psychoacoustic masking to reduce perceptibility [3]. O’Reilly et al. compared simulated and real over-the-air success for speaker-verification attacks [4]. Mathov et al. compared speaker-side and listener-side performance during Voice over IP (VoIP) calls for perturbations designed to prevent conversation-topic detection [5]. The closest comparison is TAINT, a hard-label black-box attack that combines target speech with environmental source sounds and evaluates it against four commercial ASR systems over five VoIP conferencing platforms [6]. TAINT also models VoIP channel effects and evaluates attacks under different network conditions. Its use of environmental source sounds differs from our use of source speech. In our work, we use human speech as the carrier and ask whether calibrating a local call environment improves its agreement with deployed ASR attack outcomes. Our study focuses on agreement between recording-level exact-target success rates measured under pre-deployment conditions and through deployed calls, and whether transport calibration improves that agreement. We ask how closely optimization transformations and controlled LiveKit Local calls reproduce targeted ASR attack success observed through deployed LiveKit Cloud calls. We construct an attack against white-box Whisper [7] by optimizing one perturbation for each source-target pair. Olivier and Raj demonstrated targeted white-box attacks against Whisper using optimized additive perturbations [8]. Our study examines how closely pre-deployment evaluations of such attacks reproduce success through deployed calls, and whether transport calibration improves that agreement.

2. METHOD

Given a source speech recording, our optimization adds a small perturbation to it, with the goal of forcing Whisper base.en to transcribe it to a chosen text different from the source. The optimization trains a new perturbation for every source-target pair by repeatedly making adjustments, while keeping Whisper’s weights fixed. After a perturbation candidate is selected, we construct an adversarial recording by combining the perturbation with the source. We reuse this ad-

versarial recording across the four evaluation conditions and check whether Whisper’s transcript exactly matches the target text.

We apply four separate transformation families during optimization:

1. Apply a time shift by prepending variable silence, representing timing offset.
2. Encode and decode with Opus, then apply a time shift, representing codec and timing effects.
3. Encode and decode with Opus, then apply frequency-response changes, global resampling, sample-hold concealment, and leading delay, representing measured receiver effects.
4. Encode and decode with Opus, then add a circularly shifted and scaled residual and clip the resulting sample amplitudes. The residual captures differences between a LiveKit Local received recording and an Opus-only version of the same recording.

Opus uses 20 ms frames at 48 kb/s variable bitrate in VoIP mode. It uses the exact encoded-and-decoded samples in the forward pass, while an approximate gradient is used for the backward pass.

At each optimization update, we generate 12 transformed copies: five using time shifts, three using Opus and time shifts, two using receiver effects, and two using residuals. We average Whisper’s target losses across these copies and use Adam to update the perturbation. We then enforce the perturbation constraints.

A candidate is chosen from checkpoints recorded during 7500 updates based on the highest exact-target success across 77 transformation conditions.

We used 72 distinct speakers from EARS [9], and 12 target transcripts with one shared target per six speakers. Speakers used to derive the receiver-effect settings and residuals were withheld from the 72-speaker evaluation cohort. We manually specified 12 command-like target transcripts, with two targets for each transcript length, from three to eight words. From the 72 speakers, we selected segments between 2.0 and 6.5 seconds. Next, we converted them to mono audio, and resampled to 16 kHz which we stored as PCM16. Finally, we normalized the source level to -35 dBFS RMS and appended silence to achieve a consistent duration of 7 seconds. For each speaker, we selected the first eligible segment whose Whisper transcript exactly matched its source transcript.

Modifications to the perturbation are limited to a source-relative SNR of ≥ 20 dB and a maximum change of 327 PCM16 units per sample, while the appended padding remains unchanged. These constraints limit both the perturbation’s total energy relative to the source and large changes to individual samples.

3. EXPERIMENTAL SETUP

We compare targeted ASR attack success under optimization transformations and LiveKit calls. Each evaluation uses the same adversarial recording, formed by adding the selected perturbation to the source audio. Whisper transcribes the resulting audio, and we compare its transcript with the target transcript. Fig. 1 shows the three evaluation paths.

We evaluate targeted ASR attack success under four conditions:

1. Optimization transformations: We apply the transformations used during optimization, which attempt to approximate call effects.
2. LiveKit Local baseline: The sender, LiveKit server, and receiver run on the same Linux host in separate network namespaces, with no intentionally added network impairments.
3. LiveKit Local calibrated: The same Local setup with 90 ms added delay, 4 ms normally distributed jitter, and no intentional packet loss, per configured network direction.
4. LiveKit Cloud: The Linux sender connects from the United Kingdom through LiveKit Cloud to a separate Windows receiver whose Internet traffic is routed through a Tailscale exit node in Japan.

The Local and Cloud conditions use the same application-level sender and receiver protocol and transmit the recording through actual calls. In LiveKit Local, configured network impairments alter packet delivery during the call rather than the saved waveform.

We calibrated the LiveKit Local network settings using recordings from ten EARS speakers excluded from the final evaluation cohort. Using five speakers across LiveKit Cloud and LiveKit Local calls without added impairments, we estimated the additional delay, jitter, and loss needed to reproduce the LiveKit Cloud measurements locally. We then tested nearby network settings and selected the configuration with the closest transport measurements. Validation used 25 calls per path from the remaining five speakers. The selected profile met the predefined Local–Cloud difference limits of 15 ms for median delay proxy, 3 ms for median jitter, and 8 ms for 90th-percentile jitter. It also met the packet-loss criterion: a difference of at most 0.5 percentage points or loss below 0.5% on both paths.

For each evaluation condition, we evaluate an adversarial recording and its clean source audio counterpart. For clean audio, we compare Whisper’s output with the source transcript to check whether the processing preserves the correct transcription. For the optimization transformations condition, we repeat the same 12 optimization transformations described

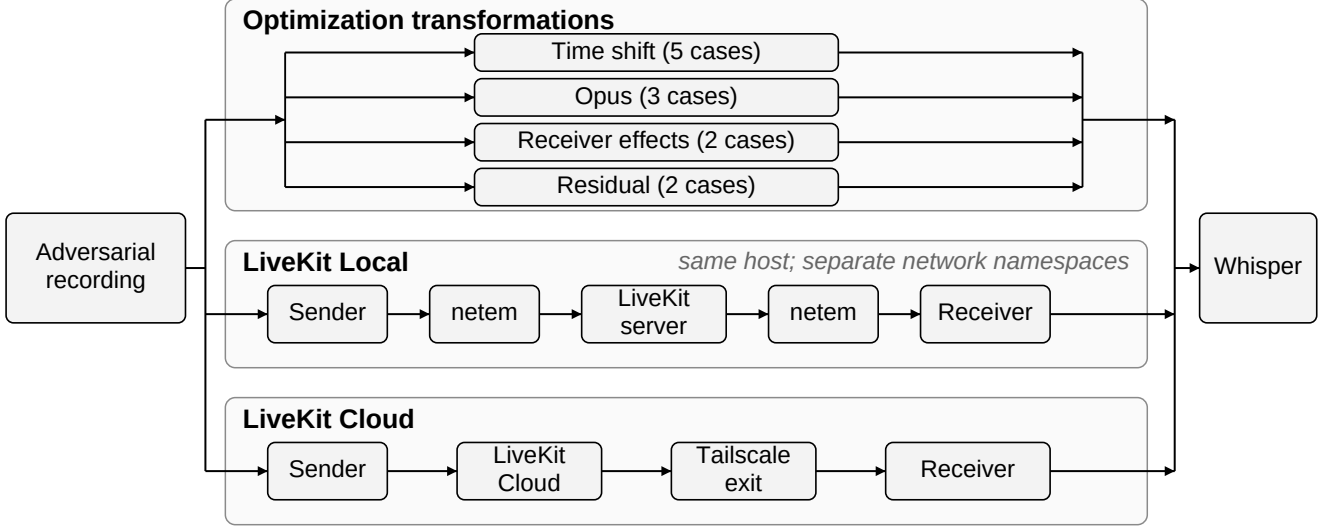


Fig. 1. Evaluation paths. The same adversarial recording passes through optimization transformations, LiveKit Local, or LiveKit Cloud before Whisper transcription.

in the Method section 32 times for each adversarial recording and its clean source counterpart. For LiveKit Local and LiveKit Cloud, we repeat each adversarial recording over ten calls and each clean recording over five calls to account for variation in delivery between calls. All evaluation conditions use Whisper base.en for English transcription with temperature 0 and previous-text conditioning disabled.

For LiveKit Local and LiveKit Cloud calls, we score the extracted 7-second payload through Whisper.

We compare transcripts disregarding capitalization, punctuation, or extra spaces, while preserving digits and apostrophes within words. An attack succeeds if the normalized transcript exactly matches the normalized target transcript.

The success rate of a recording for a given evaluation condition is determined by the number of successful tests divided by total tests. The success rate of a single evaluation condition is the average of these rates across the 72 recordings. To measure agreement, we calculate the mean absolute discrepancy between an evaluation condition and LiveKit Cloud, by averaging the absolute differences in their success rates for each of the 72 recordings.

We also calculate Spearman rank correlation with LiveKit Cloud, to measure agreement in how each evaluation condition ranks the 72 recordings by success rate. We report 95% confidence intervals using a target-stratified bootstrap with 50 000 resamples, paired across evaluation conditions. Evaluation agreements are compared using paired differences in their discrepancies and Spearman correlations with LiveKit Cloud.

4. RESULTS AND DISCUSSION

We used LiveKit Cloud as the reference for the three pre-deployment conditions. Table 1 reports the results for all four conditions.

We evaluate the same adversarial recording under four conditions: optimization transformations, LiveKit Local baseline, LiveKit Local calibrated, and LiveKit Cloud. We compare recording-level exact-target success rates with LiveKit Cloud using mean absolute discrepancy and Spearman rank correlation. We found that optimization transformations substantially overestimated LiveKit Cloud success, while the calibrated LiveKit Local condition showed the closest agreement with LiveKit Cloud among the three pre-deployment conditions. Optimization transformations produced 51.6% exact-target success compared with 21.9% through LiveKit Cloud, and their mean absolute discrepancy from LiveKit Cloud was 30.7 percentage points compared with 7.4 percentage points for the calibrated LiveKit Local condition. The results show that success under optimization transformations alone can be a poor indicator of deployment performance, and that deployment-informed LiveKit Local evaluation can provide a more representative pre-deployment test in this setup.

Compared with optimization transformations, the calibrated LiveKit Local condition had a 23.30 pp lower mean absolute discrepancy (95% CI [18.31, 28.33]) and a 0.094 higher Spearman correlation (95% CI [0.037, 0.153]), indicating both closer success rate estimates and closer ranking of recordings to LiveKit Cloud.

Calibration reduced the mean absolute discrepancy with

Table 1. Evaluation results. Brackets indicate 95% confidence intervals. “pp” indicates percentage points.

Evaluation	Mean exact-target success, %	Mean absolute discrepancy from LiveKit Cloud, pp	Spearman ρ with LiveKit Cloud	Clean exact-source transcription, %
Optimization transformations	51.6 [45.2, 58.0]	30.7 [25.9, 35.4]	0.837 [0.793, 0.869]	84.4 [82.2, 86.1]
LiveKit Local baseline	17.2 [11.2, 23.8]	11.1 [7.9, 14.7]	0.848 [0.771, 0.903]	95.8 [92.2, 98.9]
LiveKit Local calibrated	22.6 [16.1, 29.7]	7.4 [5.0, 9.9]	0.931 [0.885, 0.965]	99.2 [98.3, 100.0]
LiveKit Cloud	21.9 [16.1, 28.1]	—	—	98.1 [95.8, 99.7]

LiveKit Cloud by 3.75 pp (95% CI [0.83, 6.94]) compared with the LiveKit Local baseline.

On clean controls, LiveKit Local and LiveKit Cloud calls preserved exact source transcription in approximately 96%–99% of cases, whereas optimization transformations did so in 84.4%. This indicates that the optimization transformations were more disruptive to ordinary Whisper transcription than the call processes.

4.1. Limitations

- The study evaluates Whisper base.en, so the findings do not establish the same behavior for other ASR models.
- The evaluation uses 72 EARS recordings and 12 target transcripts, so the results are conditional on that cohort and target set.
- It evaluates one LiveKit Cloud deployment, so the calibrated LiveKit Local profile should not be treated as a universal LiveKit network profile.
- Calibration matches selected transport measurements. It does not establish that the Local and Cloud channels are equivalent in every respect.

5. CONCLUSION

We investigated how well optimization transformations and controlled LiveKit Local calls represent the LiveKit Cloud call path when evaluating targeted ASR attack success. We found that optimization transformations substantially overestimated LiveKit Cloud success, while the calibrated LiveKit Local condition provided the closest agreement with LiveKit Cloud among the three pre-deployment conditions. These results suggest that for this attack pipeline, success measured under optimization transformations alone is not sufficient evidence that an attack will remain successful after deployment. Further, calibrating LiveKit Local to match measured Cloud conditions improved agreement between Local and Cloud attack success rates. Because optimization transformations guide perturbation construction, the observed optimization-to-deployment mismatch suggests that more deployment-faithful channel models should be investigated during optimization, not only during evaluation.

Our contribution is a direct empirical comparison of optimization-time and LiveKit Local pre-deployment evaluations against deployed targeted ASR behavior.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study used only existing recordings from the EARS dataset [9]. No new participants were recruited or human-subject recordings collected. Ethical approval was not required for this study.

7. ACKNOWLEDGEMENTS

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